What is a virtual environment?

A virtual environment is a self-contained directory that houses a specific version of Python along with its own set of packages and libraries. It isolates your project's dependencies from your global Python installation and other projects. This ensures that upgrades or changes in one project won't break another.

Installing Python

Windows

1. Download the official installer from the Python website.
2. Run the installer and **crucially check the box** that says **"Add python.exe to PATH"** at the bottom of the window.
3. Click **Install Now**.
4. Verify the installation by opening Command Prompt and running:

```
python --version
```

Creating and Activating a Virtual Environment

Windows (Command Prompt)

1. Navigate to your project folder:

```
cd path\to\your\project
```

2. Create the virtual environment (replace `myenv` with your preferred environment name):

```
python -m venv myenv
```

3. Activate the virtual environment:

```
myenv\Scripts\activate.bat
```

4. *(Your terminal prompt will change to show `(myenv)`).*

3. Deactivating the Virtual Environment

When you are done working on your project, you can exit the virtual environment by running:

```
deactivate
```